Midterm 2 Review

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Course Syllabus and reading assignments:

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b. Stability analysis
c. Steady-state errors
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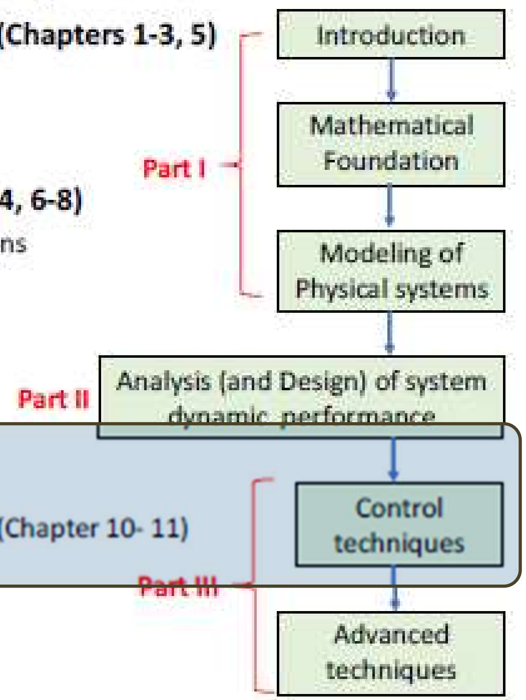
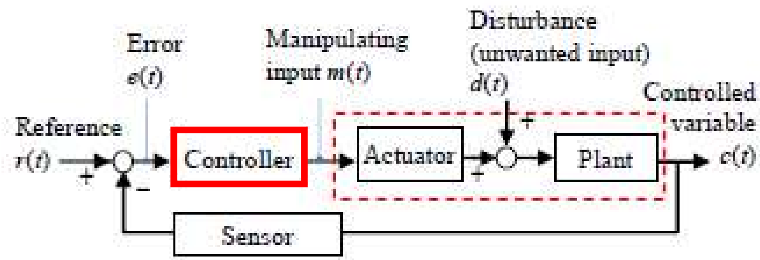
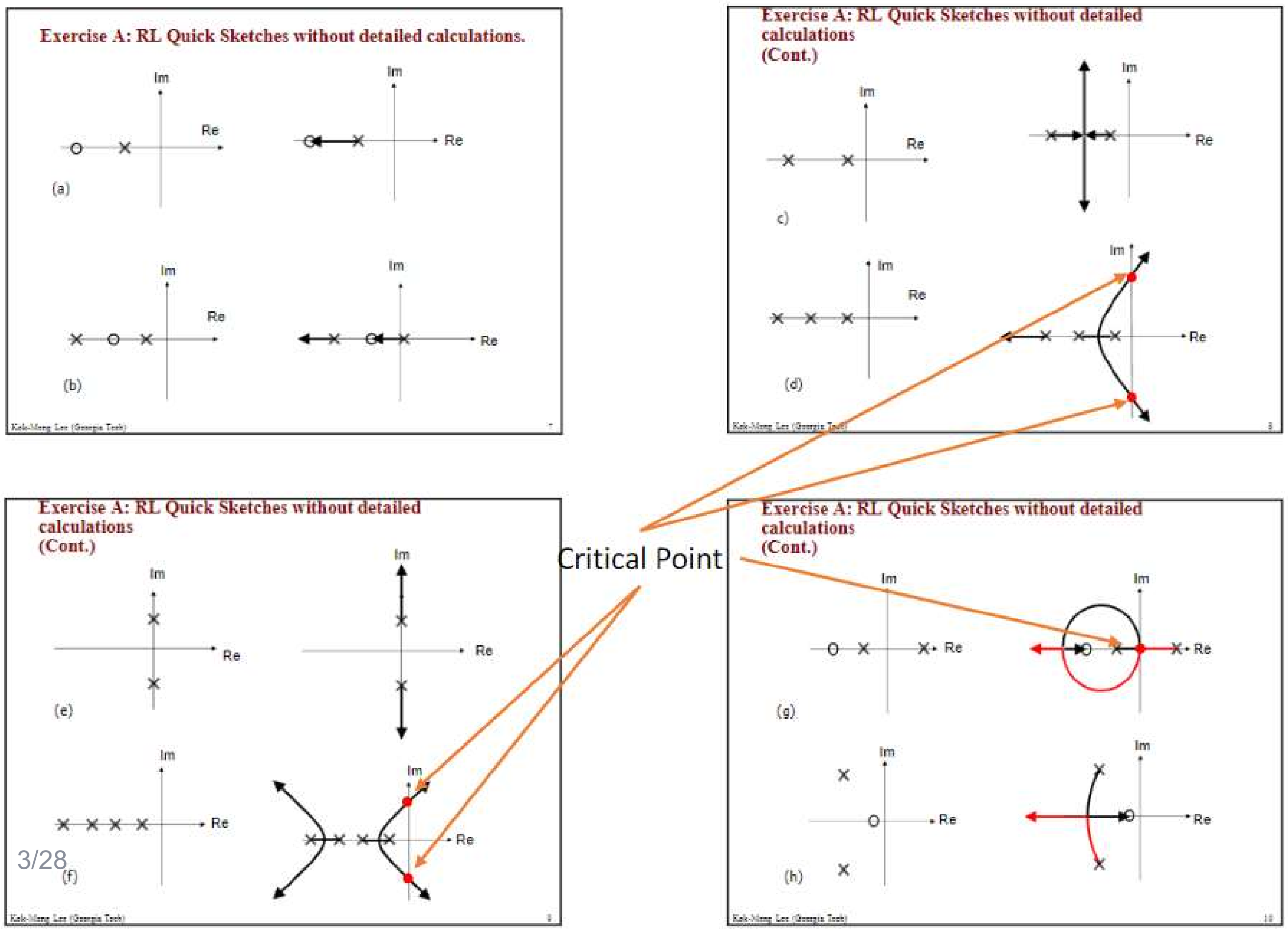




SISO Automatic (Closed-loop) Control System

Control objective: $c(t) \to r(t)$ regardless of $d(t)$.

Root Locus Design



Points on RL satisfy two conditions:

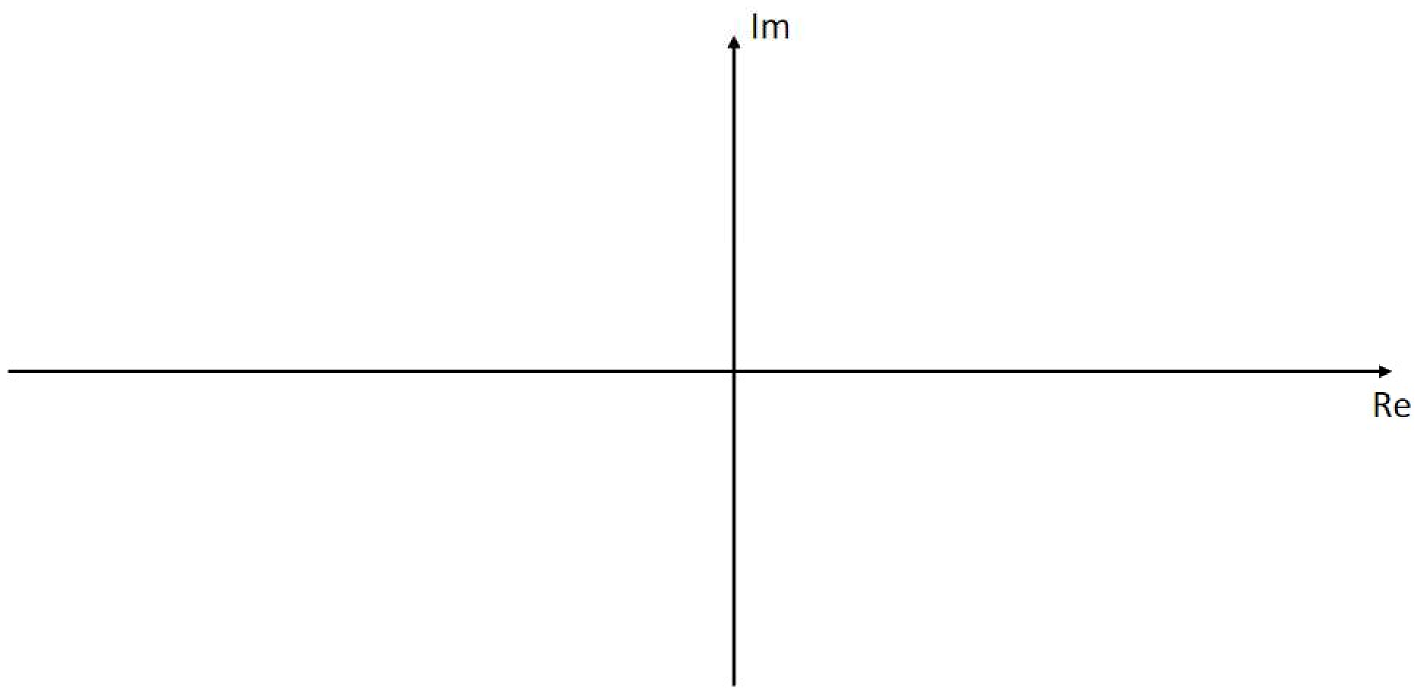
Mag. $K\left|\frac{N(s)}{D(s)}\right|=1 \Rightarrow K=\frac{|D(s)|}{|N(s)|}$

Phase $\angle\left[\frac{N(s)}{D(s)}\right]=\begin{cases}\pm 180(2l+1) & K \ge 0\\ \pm 180(2l) & K<0\end{cases}$ where $l=0,1,2,\cdots$

$$\sum \frac{1}{\sigma_b - p_i} = \sum \frac{1}{\sigma_b - z_i}$$

Example

$$G(s) = \frac{s+2}{(s-2)(s+1)}, \sigma_b = 4$$



Example

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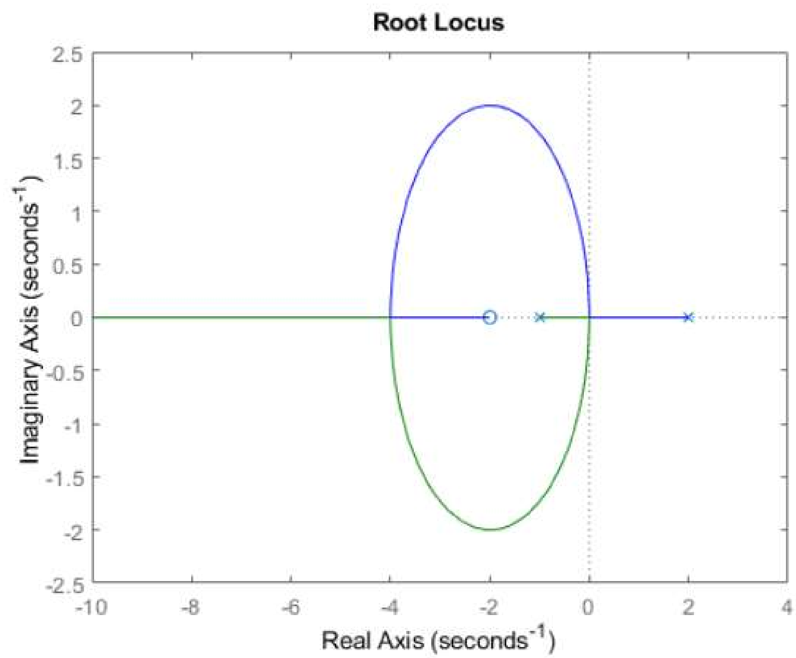

Bode Diagram

2 plots on logarithmic scales

Magnitude	$20\log_{10}\lvert G(j\omega)\rvert$ dB
Phase	$\angle G(j\omega)$ degrees

Standard Forms

1st order lead (+), PD controller (High-pass filter):	$Ts+1$	$G(s)=(Ts+1)^{\pm 1}$
1st order lag (−), Mech damper (Low-pass filter):	$1/(Ts+1)$	

$T\omega_c = 1$

2nd order lead (+) and lag (−) : Example: mass-spring-damper system.	$G(s)=\left(\frac{s^2}{\omega_n^2}+2\varsigma\frac{s}{\omega_n}+1\right)^{\pm 1}$

Routh Criterion Vs. Root Locus – CL poles' place in real-imaginary plane
Bode Plot - IO magnitude ratio(gain) & phase shift
Nyquist – IO magnitude ratio(gain) & phase shift in real-imaginary plane

Close loop Open loop

"Critical gain from each criterion is the same"

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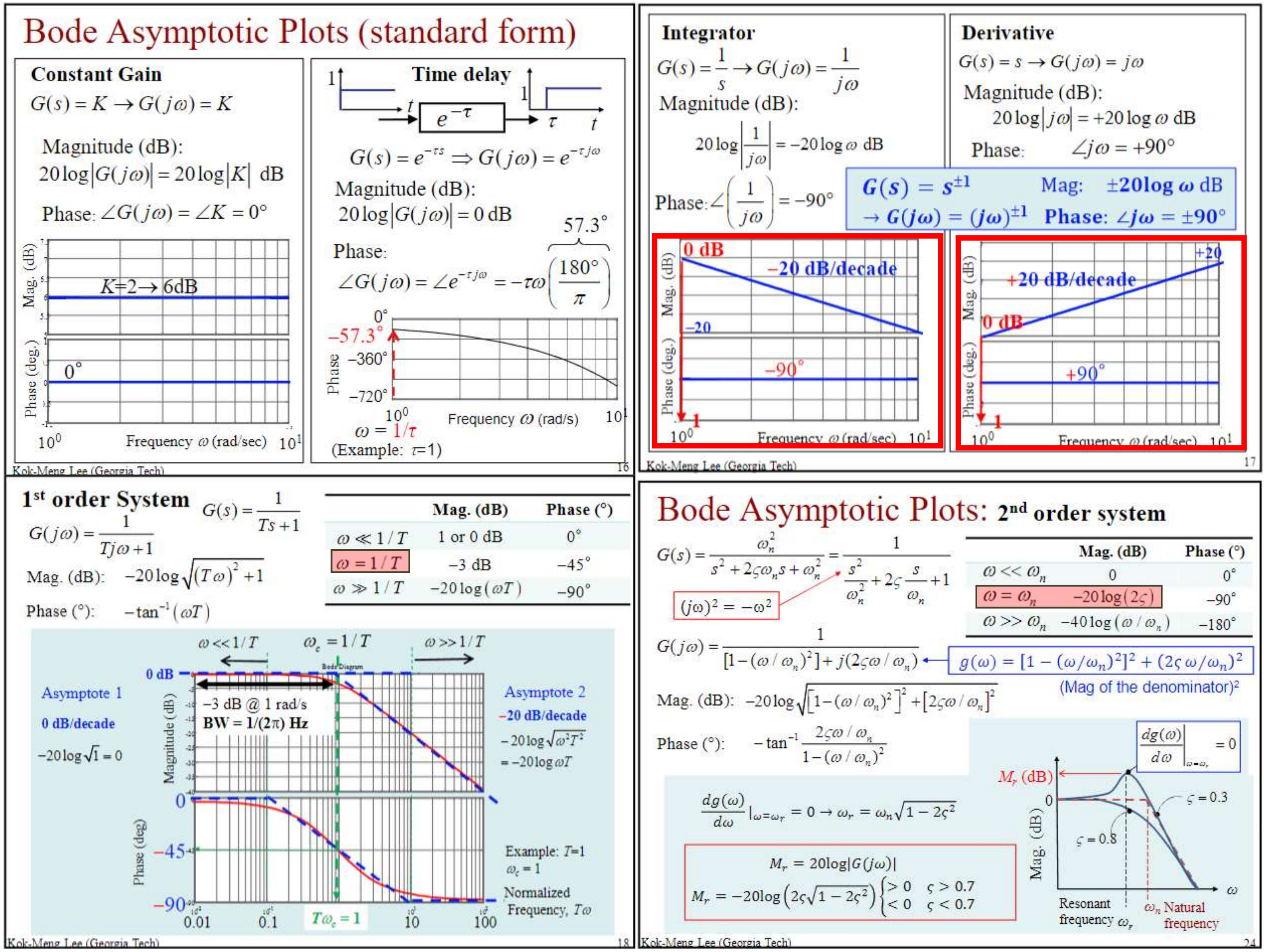
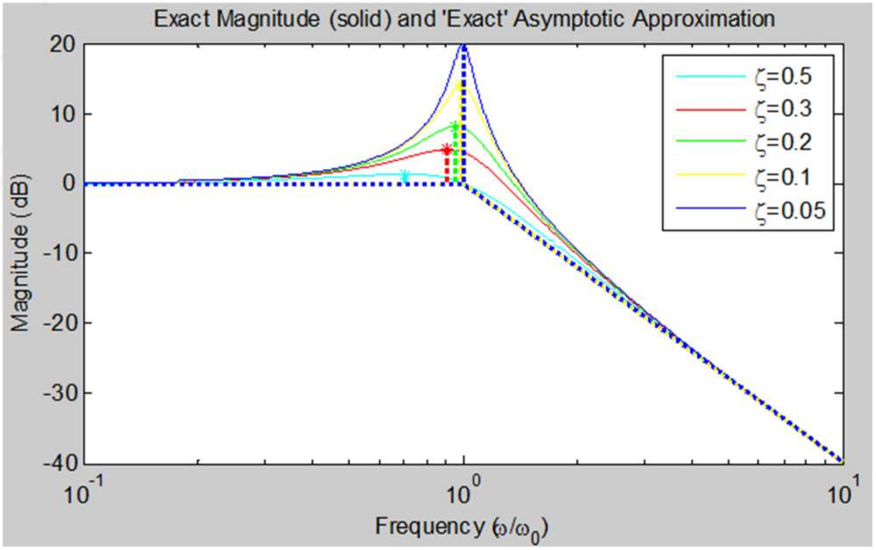
Bode Diagram

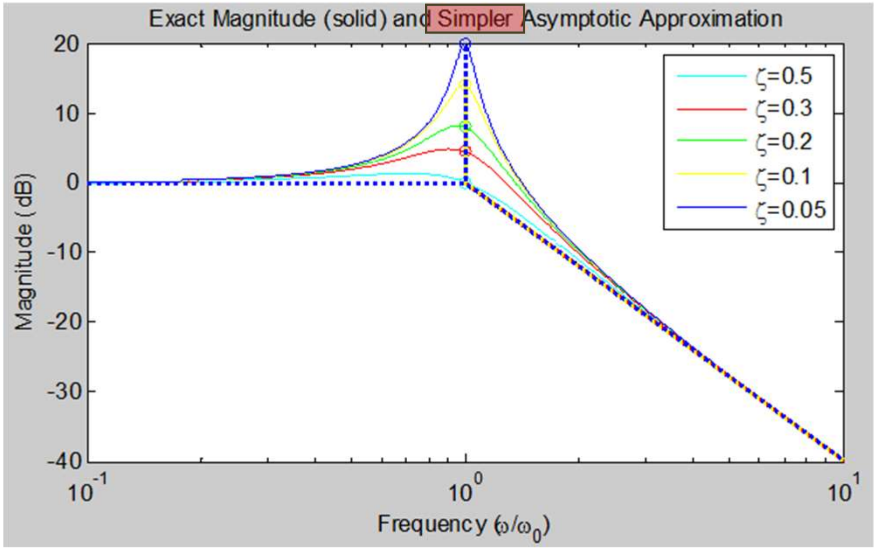


Position of Curve:

$\frac{1}{10\omega_c}$ & $10\omega_c$

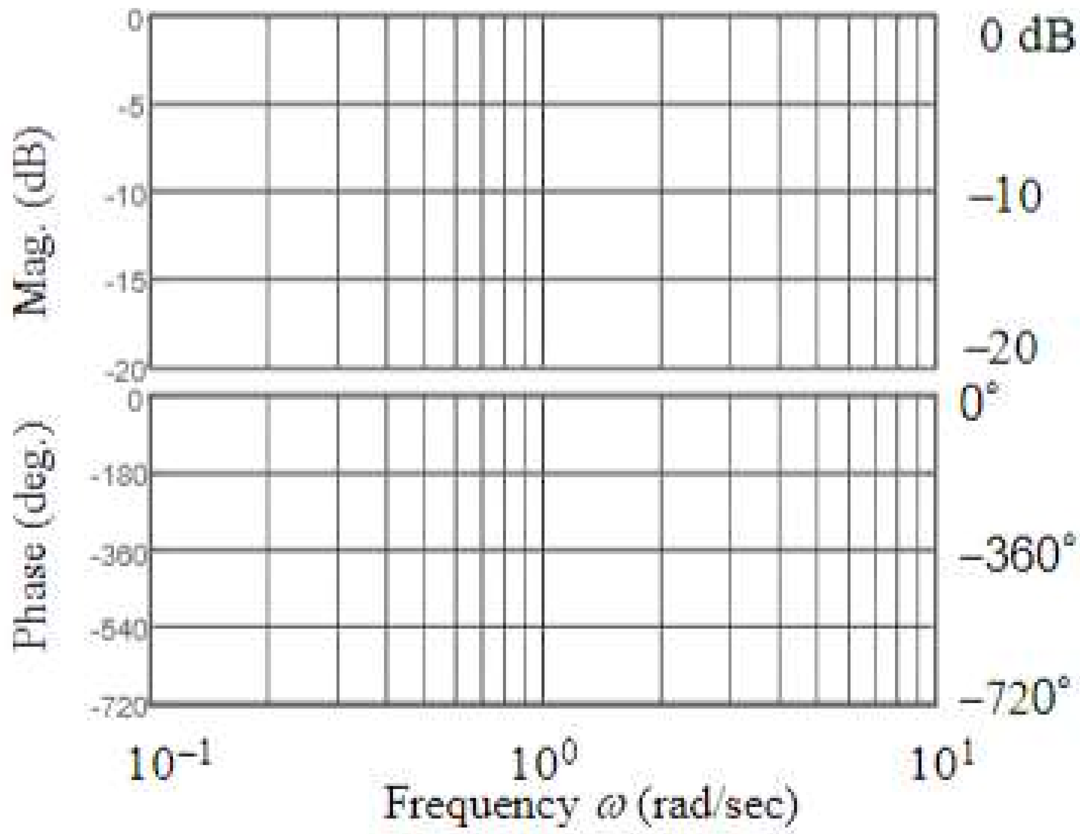
Bode Diagram





Using simpler approximation is okay

Example - $G(s) = \frac{1}{s+1}$



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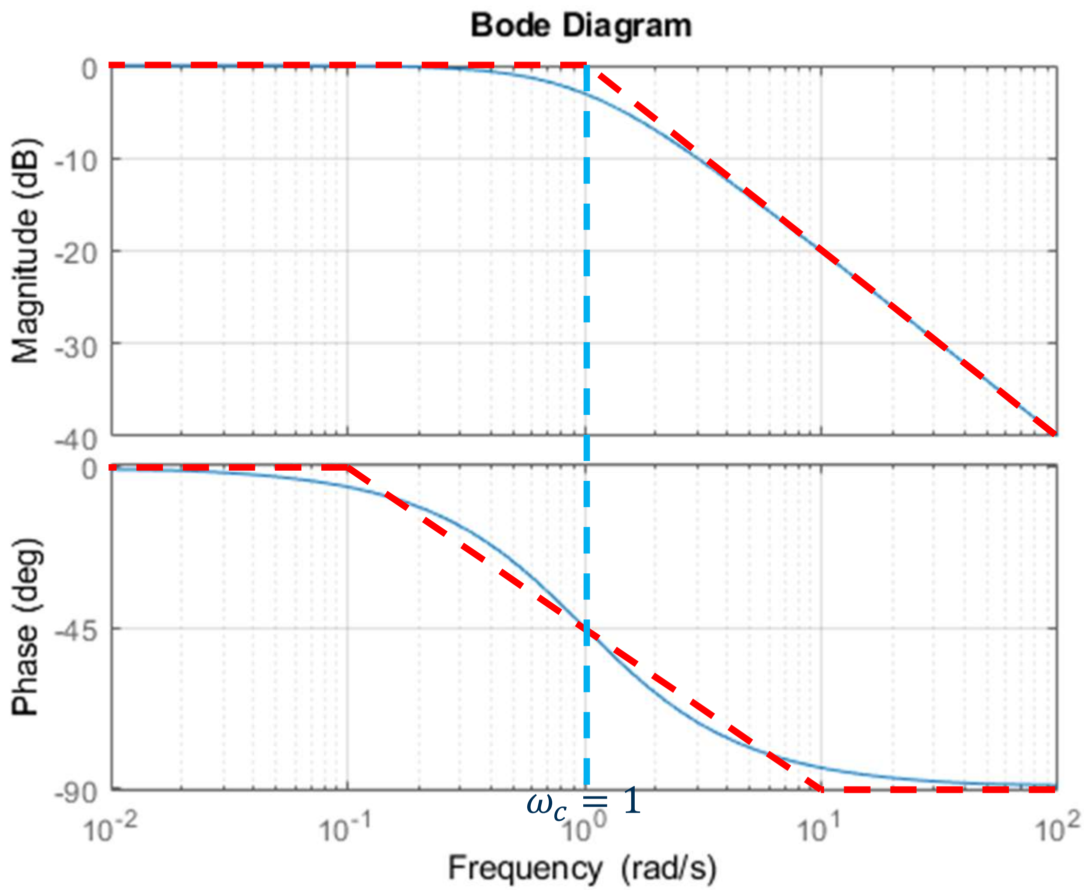
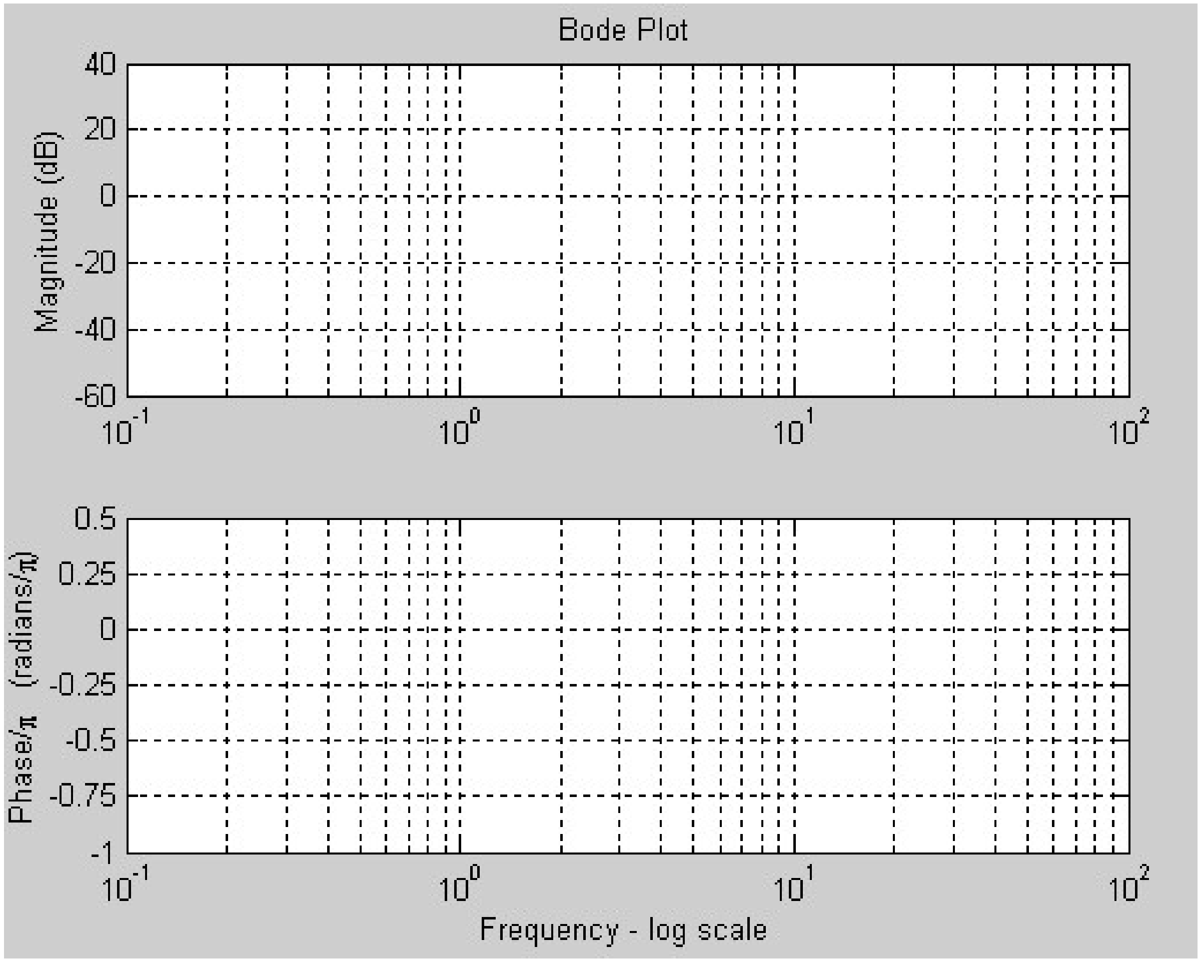
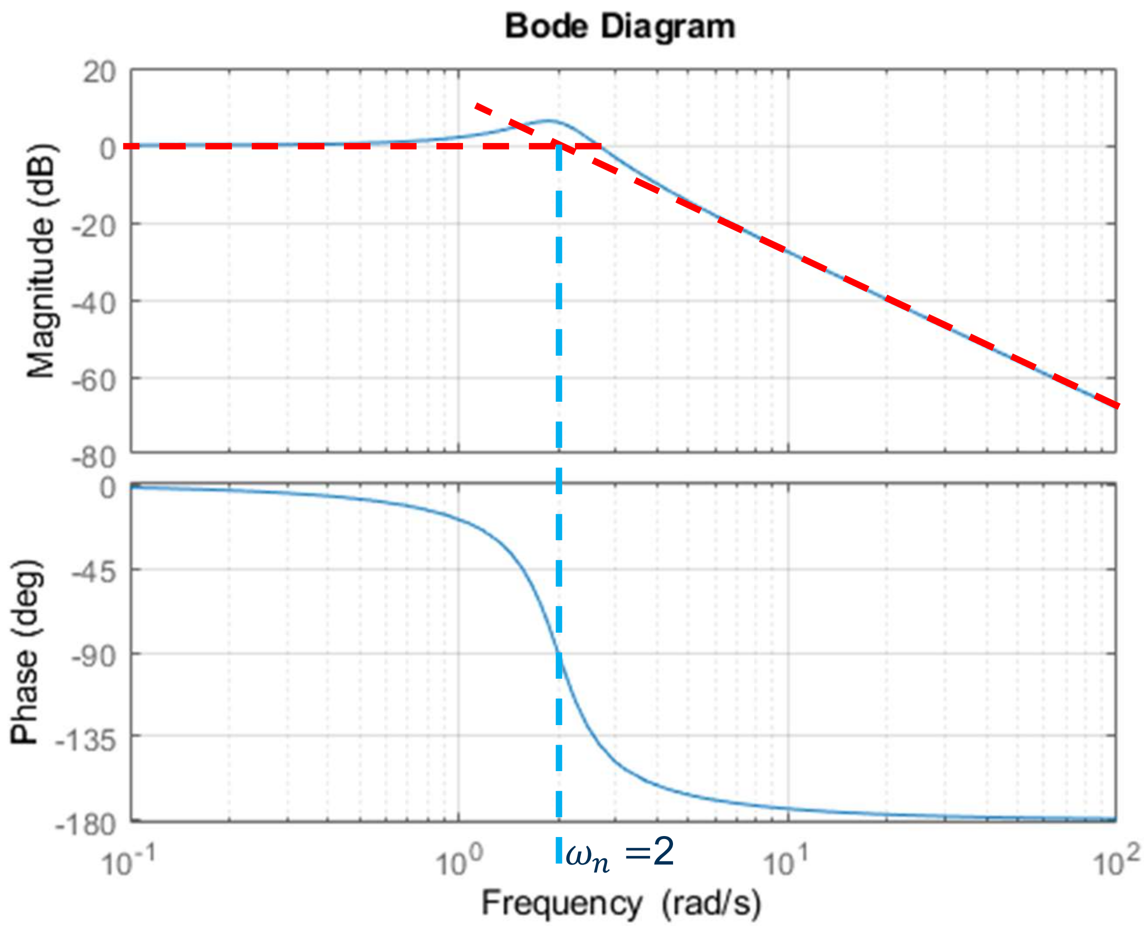
Example - $G(s) = \frac{1}{s+1}$



Example - $G(s) = \frac{4}{s^2+s+4}$

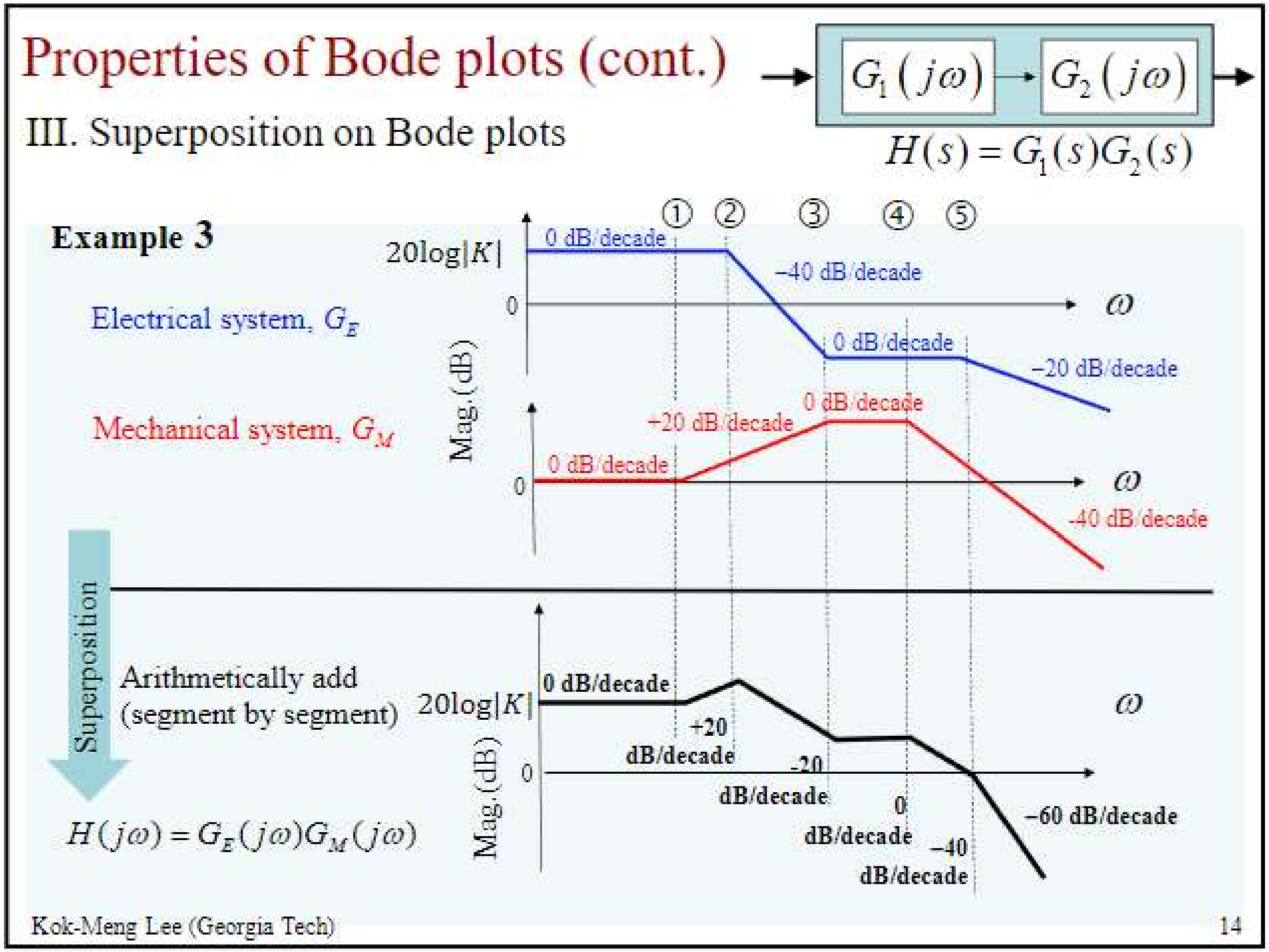


Example - $G(s) = \frac{4}{s^2+s+4}$

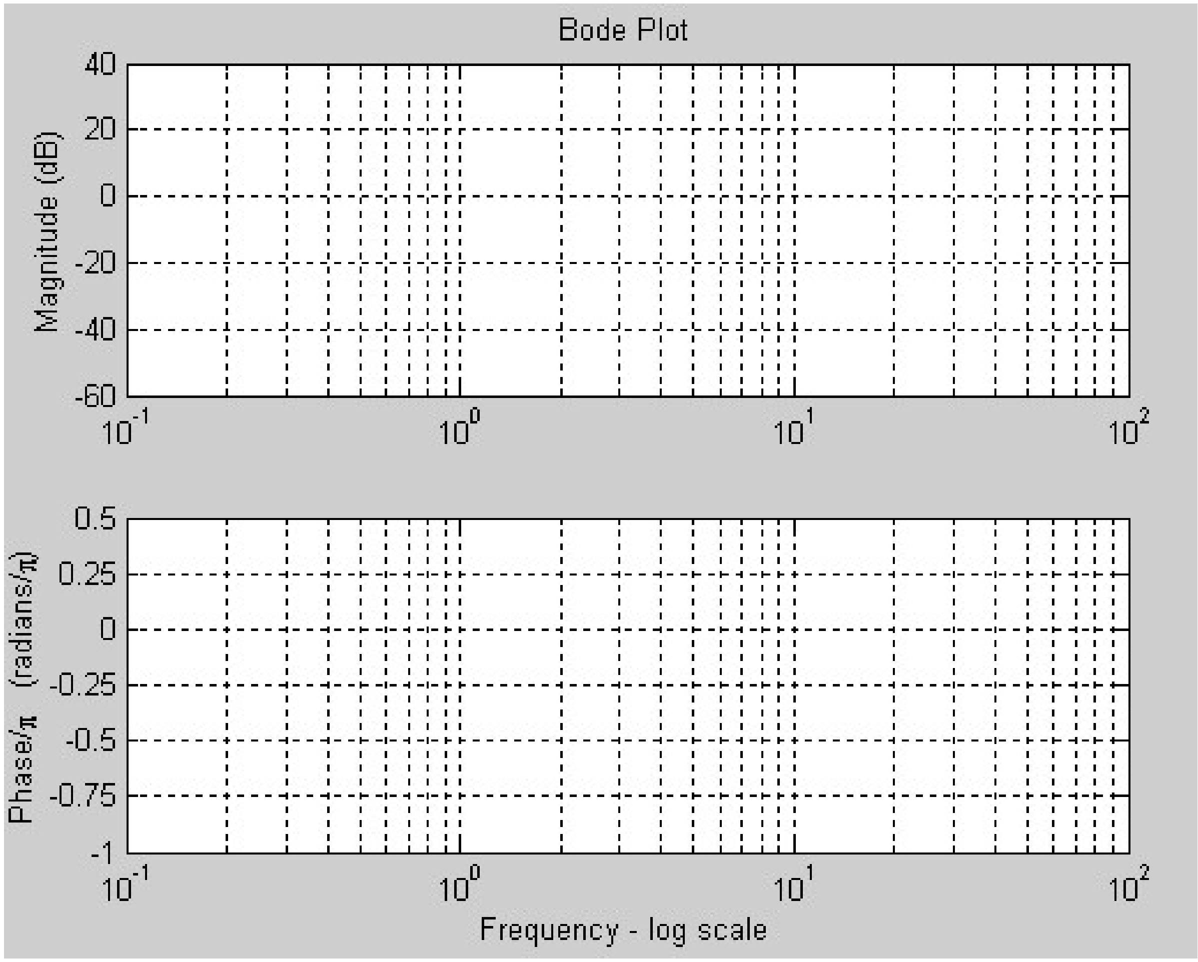


Bode Diagram - Superposition

https://github.com/echeever/BodePlotGui



Bode Diagram - $G(s) = \dfrac{10}{s(s+1)(0.5s+1)}$



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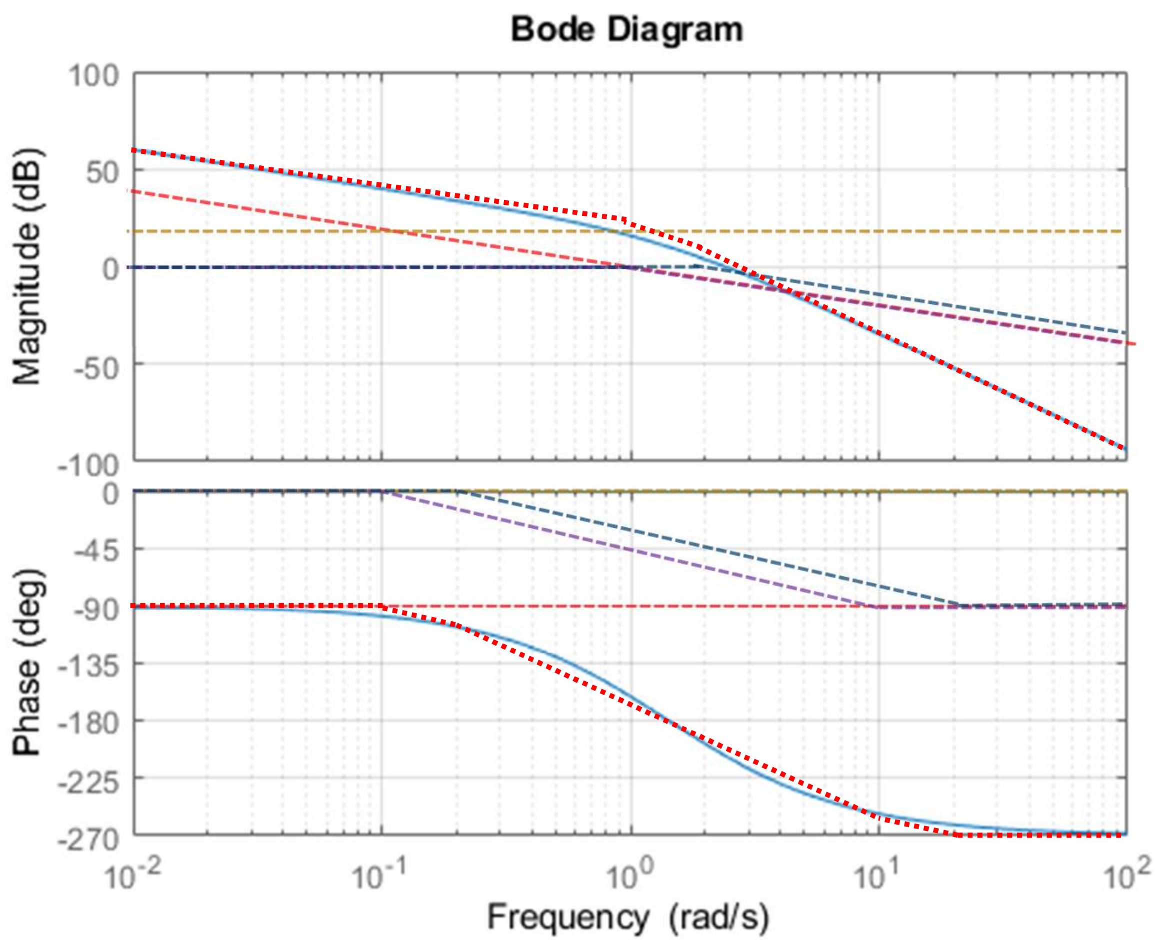


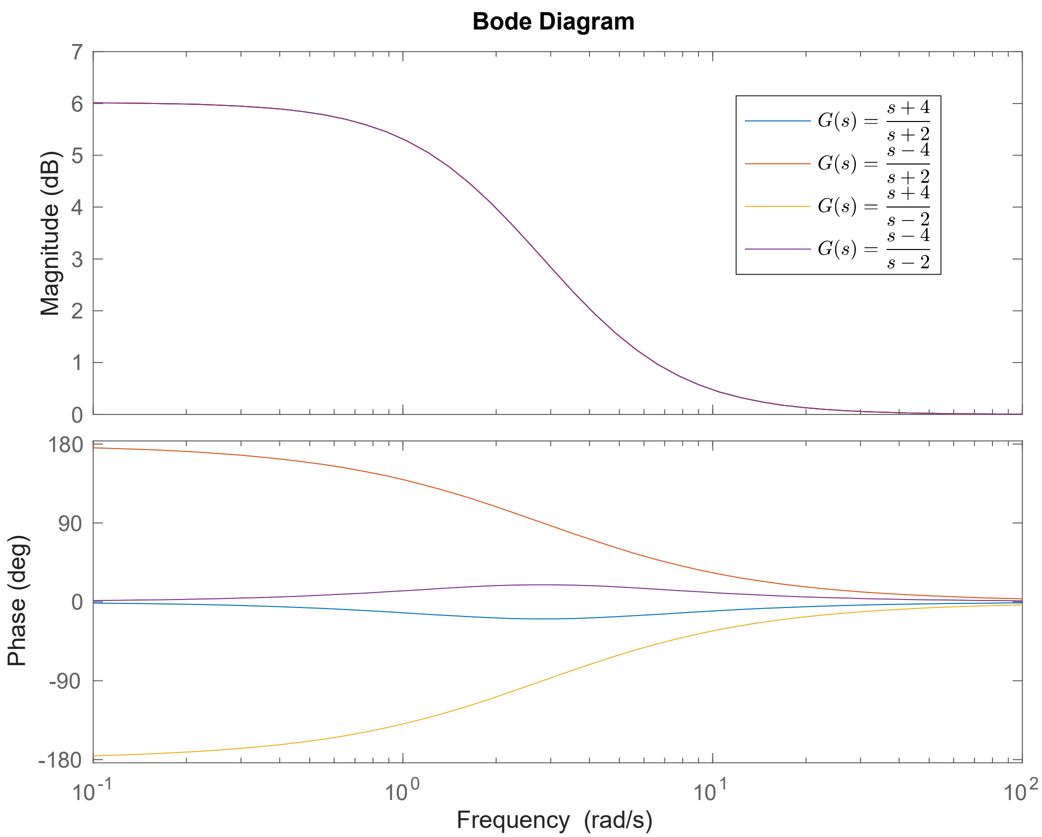
$K = 10$

$G(s) = \dfrac{1}{s}$

$G(s) = \dfrac{1}{s+1}$

$G(s) = \dfrac{1}{0.5s+1}$

Bode Diagram – Effect of Switching Sign: ex) $G(s) = \frac{s \pm 4}{s \pm 2}$



1. Sign does not affect the magnitude
2. **But it affects phase**

Lag Compensator

Position $K_{pos} = \lim_{s\to 0} G(s) = K$

Velocity $K_{vel} = \lim_{s\to 0} sG(s)$

Acceleration $K_{acc} = \lim_{s\to 0} s^2 G(s)$

Error constant (Bode Mag. Plot)

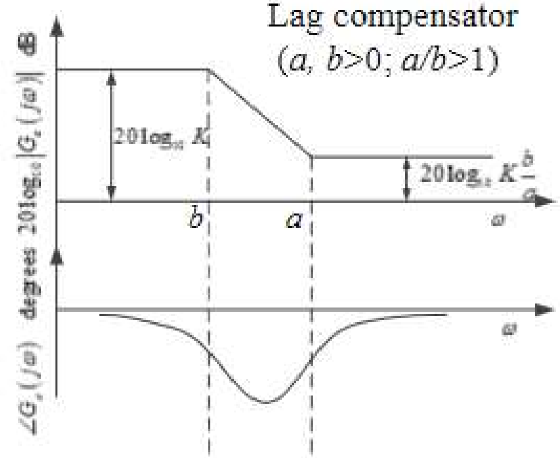
$$20\log\left|K\left(j\omega\right)^{-N}\right| = 20\log K_i$$

where $i = pos,\ vel,$ or acc

Lag Compensator (Design procedure)

Compensator: $G_c\left(s\right) = K\dfrac{s/a+1}{s/b+1} = \left(K\dfrac{b}{a}\right)\dfrac{s+a}{s+b}$

To meet an error constant spec but keep phase margin and bandwidth constant:



Step 1: Determine the required increase in Bode gain and set the compensator gain K to this value.

Step 2: Let the frequency ratio equal to K.
$\dfrac{a}{b} = K;\ K\dfrac{b}{a} = 1$ at high frequency.

Step 3: Choose the "zero" frequency "a" at least a decade below the cross-over frequency ω_c to avoid phase lag contribution at ω_c $\left(a \le 0.1\omega_c\right)$

Step 4: The frequencies "b" can be found from Steps 2 and 3: $b = \underbrace{\left(b/a\right)}_{1/K}a = a/K$

$$G_c\left(s\right) = K\frac{10s/\omega_c+1}{10Ks/\omega_c+1}$$

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Lead Compensator

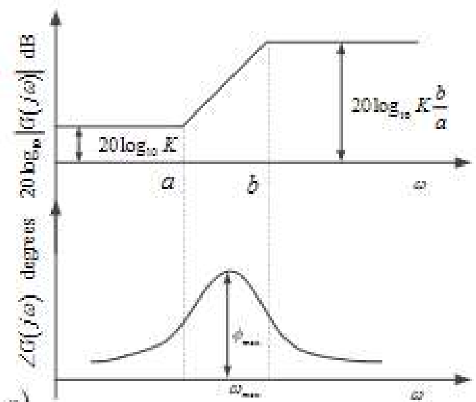
$G_c(s) = K\dfrac{s/a+1}{s/b+1} = \left(K\dfrac{b}{a}\right)\dfrac{s+a}{s+b}$

$|G(j\omega)| = |G(s)| = 1$

Lead Compensator (Design procedure)

To meet bandwidth and phase margin specs with an unspecified change in gain (error constant).

Lead compensator (*a, b>0; a/b<1*)



Maximum phase lead ϕ_{max} , $\sin\phi_{max} = \dfrac{b/a-1}{b/a+1}$

Frequency for maximum lead, $\omega_{max}^2 = ab$

Gain of $\dfrac{s/a+1}{s/b+1}$ at $\omega = \omega_{max}$, $20\log_{10}|G(j\omega_{max})| = 10\ \log_{10}(b/a)$

Step 1: Draw the open-loop Bode plots of the uncompensated system.

Step 2: Determine the phase lead required at the specified cross-over frequency to meet the phase margin spec.

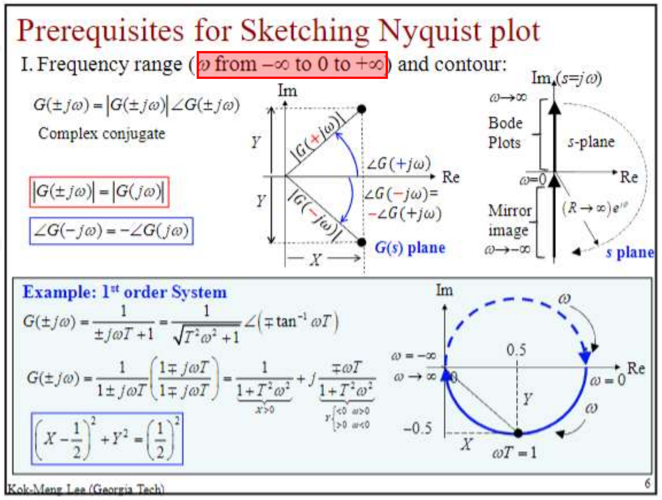
Step 3: Let thus required lead angle be ϕ_{max} and solve (2) for b/a.

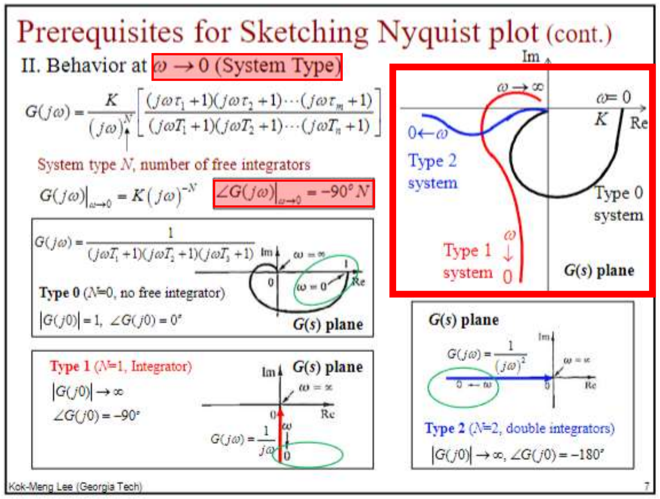
Step 4: Let this maximum lead occur at ω_c so that (3) becomes $\omega_{max}^2 = \omega_c^2 = ab$. Use this equation and the result of Step 3 to find "a" and "b":

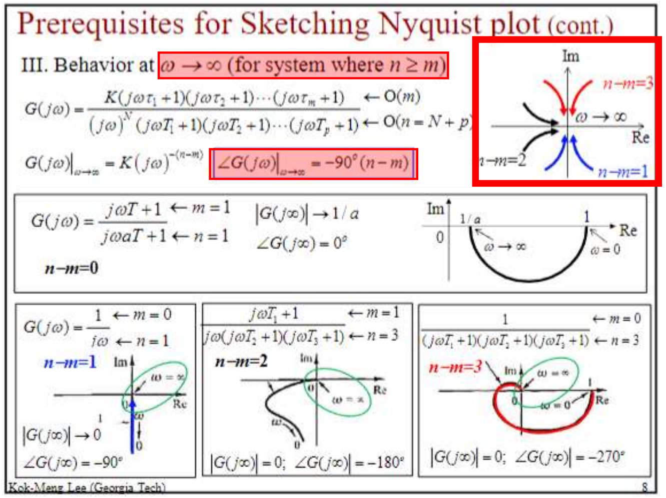
Step 5: Graphically add the frequency response of the lead compensator to the Bode magnitude portion of the uncompensated system.

Step 6: Adjust the gain K of the compensator until the magnitude of the compensated system goes through 0 dB at $\omega = \omega_c$.

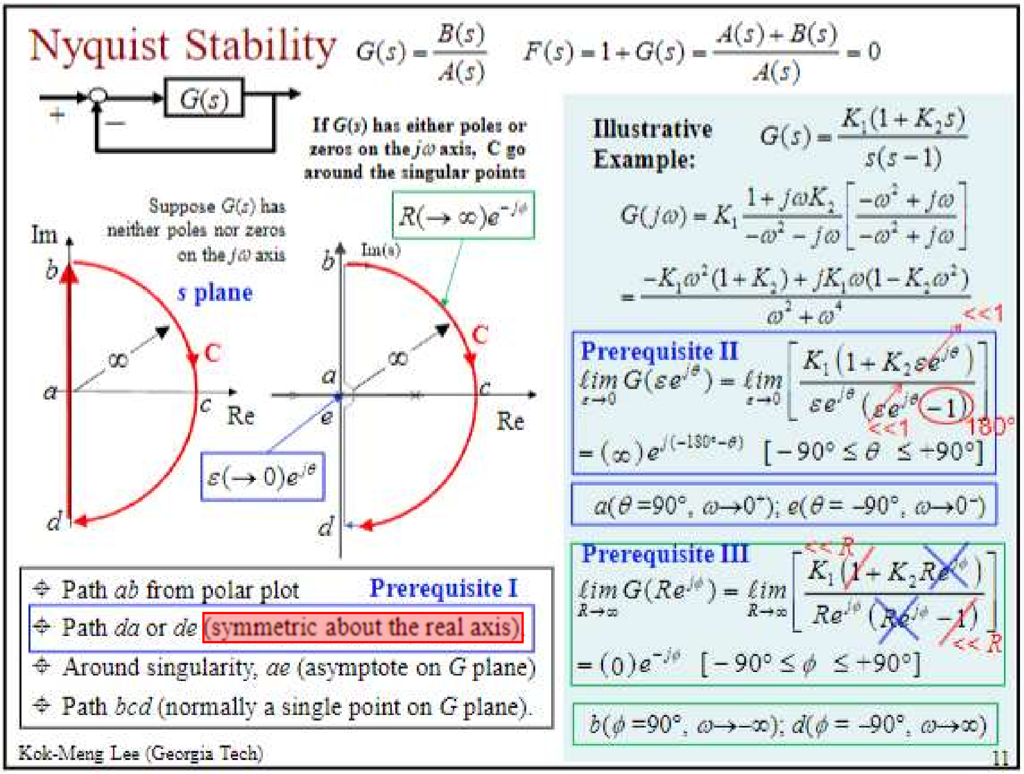
Sketching Nyquist Plot

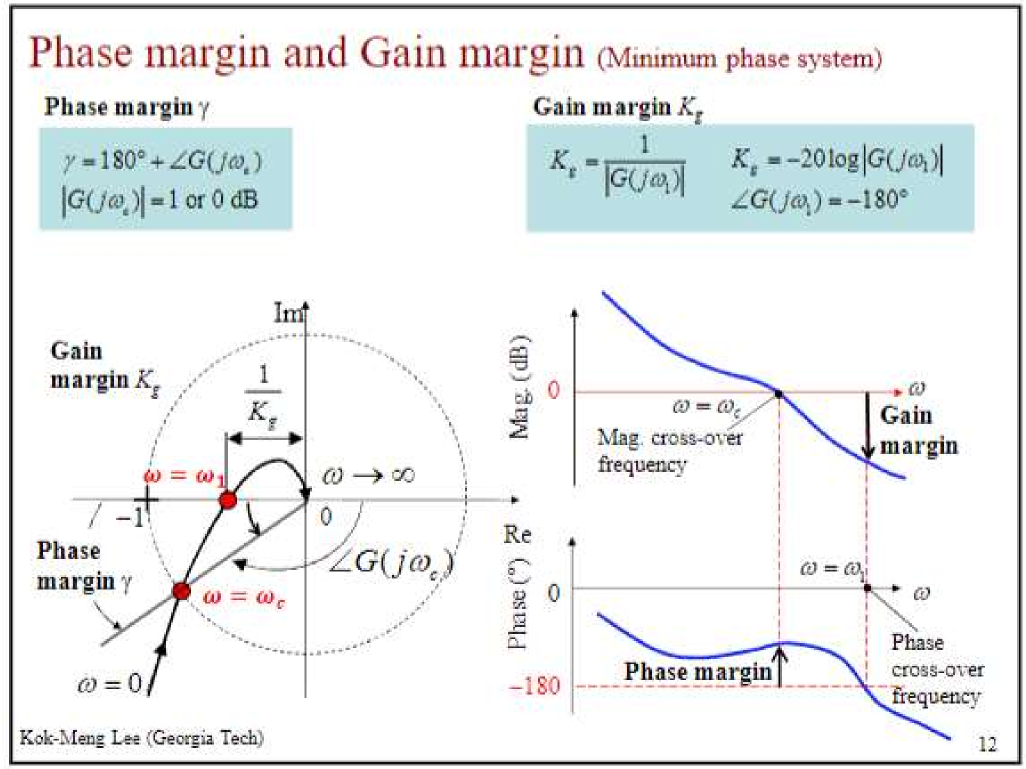






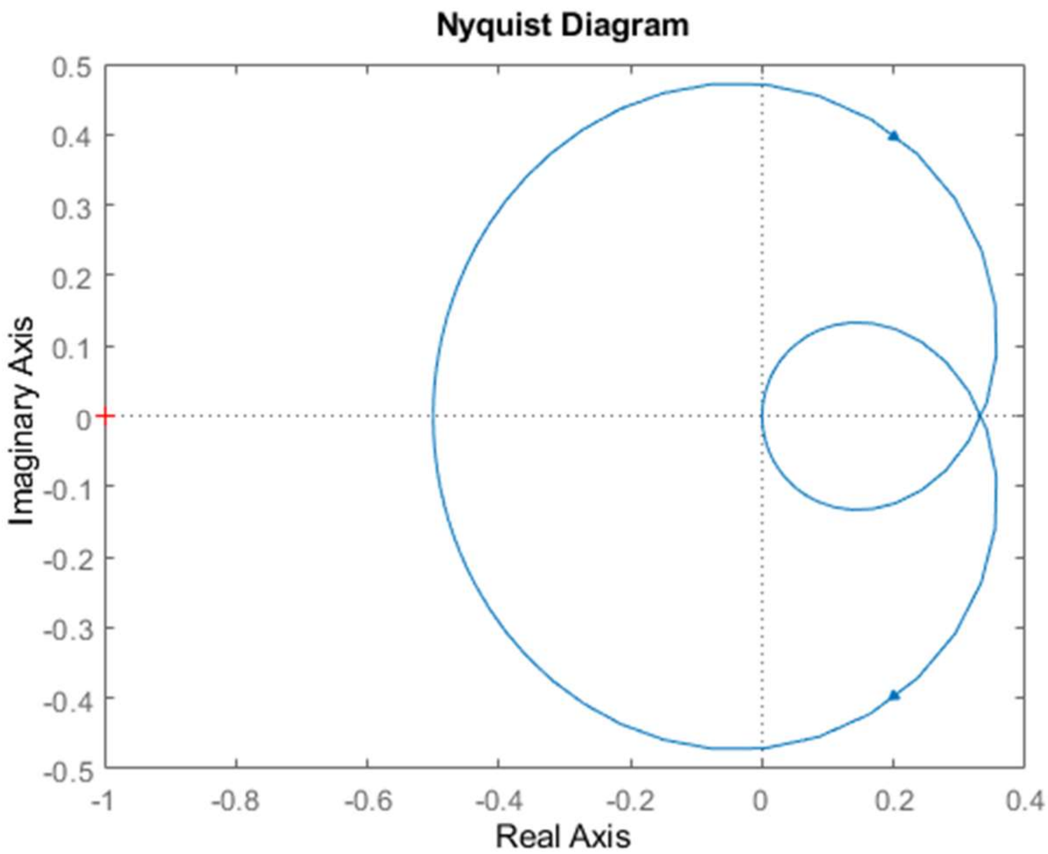
Sketching Nyquist Plot





Example - $G(s) = \frac{s-1}{(s+2)(s+1)} = \frac{4\omega^2-2}{\omega^4+5\omega^2+4} - \frac{j\omega(\omega-5)}{\omega^4+5\omega^2+4}$

Example - $G(s) = \frac{s-1}{(s+2)(s+1)} = \frac{4\omega^2-2}{\omega^4+5\omega^2+4} - \frac{j\omega(\omega-5)}{\omega^4+5\omega^2+4}$



Nyquist Criteria - $G(s) = \frac{s-1}{(s+2)(s+1)}$

Nyquist Stability theorem:

$N = Z - P \Rightarrow Z = N + P$

P=# of **unstable open-loop poles** [or poles of $F(s)$]

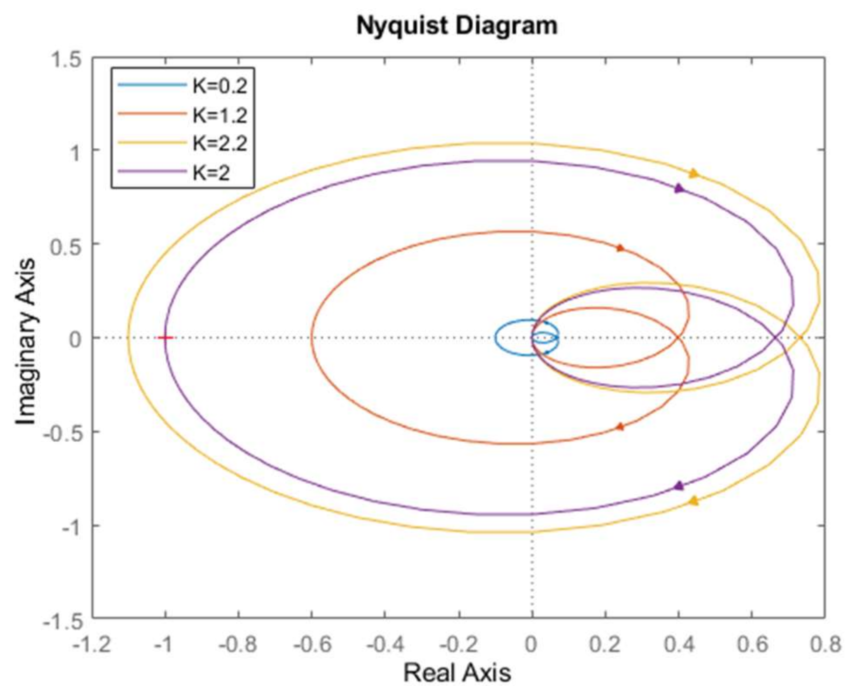
Z=# of **unstable closed-loop poles** [or zeros of $F(s)$]

N=# of **clockwise** rotations of $F(s)$ around the origin of the $F(s)$ plane

= # of clockwise rotations of $G(s)$ around $(-1, j0)$.

1) This is normally used to find Z based on P and N.

2) **For stability, Z=0.**



Verify whether Z changes from 0 to another value.

State Space

System matrix, $n \times n$ Input matrix, $n \times r$

State equation: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$

Output equation: $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$

Output matrix, $m \times n$ Direct transmission matrix , $m \times r$

$\mathbf{x} = n \times 1$ State vector

$\mathbf{u} = r \times 1$ Input vector

$\mathbf{y} = m \times 1$ Output vector

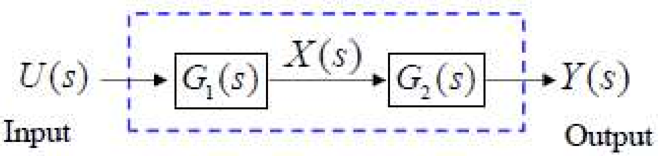
Transfer function to State Space

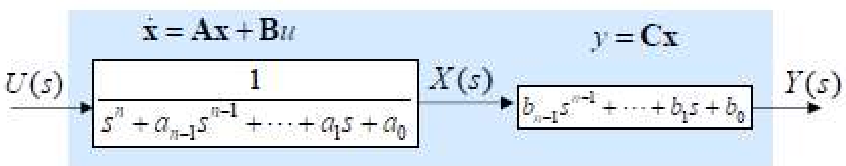
From transfer function (SISO) to state-space equation

$$G(s)=\frac{b_{n-1}s^{n-1}+\cdots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}=\frac{Y(s)}{U(s)}=\underbrace{\frac{X(s)}{U(s)}}_{G_1(s)}\underbrace{\frac{Y(s)}{X(s)}}_{G_2(s)}$$

Let $\dfrac{X(s)}{U(s)}=\dfrac{1}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$

$$\frac{Y(s)}{X(s)}=b_{n-1}s^{n-1}+\cdots+b_1s+b_0$$





State-space representation (SISO)

$$\frac{X(s)}{U(s)}=\frac{1}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$$

$$\frac{d^nx}{dt^n}+a_{n-1}\frac{d^{n-1}x}{dt^{n-1}}+\cdots+a_1\frac{dx}{dt}+a_0x=u$$

Let

State vector

$$\begin{bmatrix} x_1 = x \\ x_2 = \dfrac{dx}{dt}=\dot{x}_1 \\ x_3 = \dfrac{d^2x}{dt^2}=\dot{x}_2 \\ \vdots \\ x_n = \dfrac{d^{n-1}x}{dt^{n-1}}=\dot{x}_{n-1} \end{bmatrix}$$

$$\underbrace{\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \vdots\\ \dot{x}_{n-1}\\ \dot{x}_n\end{bmatrix}=\begin{bmatrix}0 & 1 & 0\cdots & 0\cdots 0\\ 0 & 0 & 1 & 0\cdots 0\\ \vdots & & & \\ 0 & 0 & 0 & 0\cdots 1\\ -a_0 & -a_1 & -a_2\cdots & \cdots -a_{n-1}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ \vdots\\ x_{n-1}\\ x_n\end{bmatrix}+\begin{bmatrix}0\\0\\0\\0\\1\end{bmatrix}u}$$

State equation: $\boxed{\dot{\mathbf{x}}=\mathbf{A}\mathbf{x}+\mathbf{B}u}$

With zero initial condition:

$$\frac{dx(t)}{dt}\leftrightarrow sX_1(s)$$

$$\frac{d^2x(t)}{dt^2}\leftrightarrow s^2X_1(s)$$

$$\frac{d^nx(t)}{dt^n}\leftrightarrow s^nX_1(s)$$

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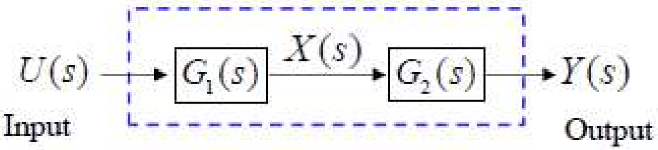
Transfer function to State Space

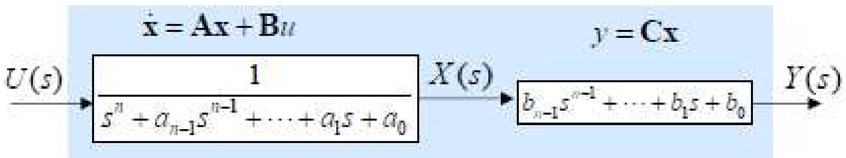
From transfer function (SISO) to state-space equation

$$G(s)=\frac{b_{n-1}s^{n-1}+\cdots+b_1s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}=\frac{Y(s)}{U(s)}=\underbrace{\frac{X(s)}{U(s)}}_{G_1(s)}\underbrace{\frac{Y(s)}{X(s)}}_{G_2(s)}$$

Let $\dfrac{X(s)}{U(s)}=\dfrac{1}{s^n+a_{n-1}s^{n-1}+\cdots+a_1s+a_0}$

$$\frac{Y(s)}{X(s)}=b_{n-1}s^{n-1}+\cdots+b_1s+b_0$$





State-space representation (SISO)

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State equation: $\boxed{\dot{\mathbf{x}}=\mathbf{A}\mathbf{x}+\mathbf{B}u}$

With zero initial condition,

$$\frac{dx(t)}{dt}\leftrightarrow sX_I(s)$$

$$\frac{d^2x(t)}{dt^2}\leftrightarrow s^2X_I(s)$$

$$\frac{d^nx(t)}{dt^n}\leftrightarrow s^nX_I(s)$$

State Space to Transfer Function

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$
$$y = \mathbf{C}\mathbf{x}$$

$$G(s) = \frac{Y(s)}{U(s)} = \underset{1\times n}{\mathbf{C}}(s\mathbf{I} - \underset{n\times n}{\mathbf{A}})^{-1}\underset{n\times 1}{\mathbf{B}}$$

Characteristic equation: det [sI − A] = 0

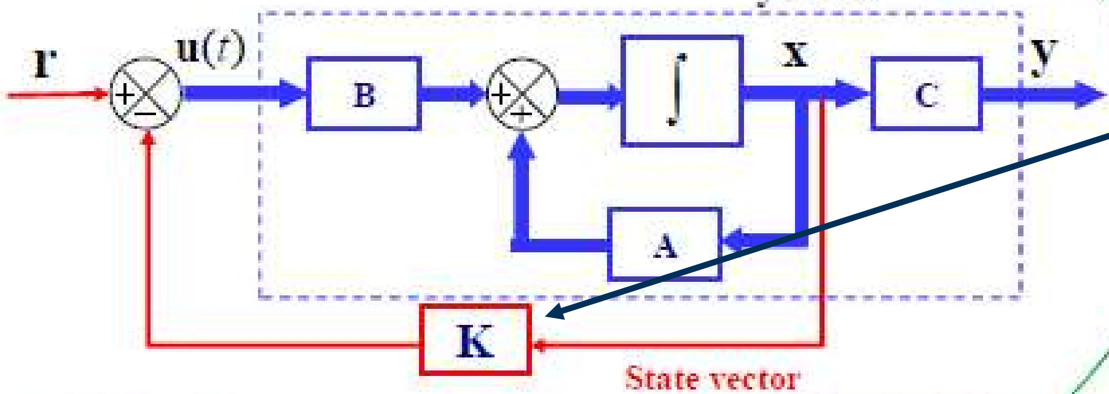
Full State Feedback Control

Full State feedback control (pole placement)

Assume all state variables are measurable.

Plant: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$

$\mathbf{y} = \mathbf{C}\mathbf{x}$



Gain may not be constant

- Closed-loop system

Control law: $\mathbf{u}(t) = -\mathbf{K}\mathbf{x}(t) + \mathbf{r}$

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x} + \mathbf{B}\mathbf{r}$$

Closed loop characteristic equation: $\left| s\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K}) \right| = 0$

Desired closed-loop poles can be placed using pole placement method.